

Appendix D: Helix Hopper Vibrator

Some materials require intermittent vibration to help the material flow. For this reason, Hapman offers electric and pneumatic vibrator options. Existing units can also be retrofitted with a vibrator if requested.



Appendix D: Helix Hopper Vibrator

NOTICE

Running the vibrator with no material in the hopper can lead to unwarranted damage to the hopper.

NOTICE

Vibrators should be run intermittently. Any damage occurring from running the vibrator continuously will not be warranted.

1.

The electric vibrator will be controlled by a cycle timer relay inside the Hapman control panel. It requires the Helix™ to be running for the Vibrator to turn on.

2.

Once the Helix™ is running, power is supplied to the A1 and 15 contacts on the relay timer.

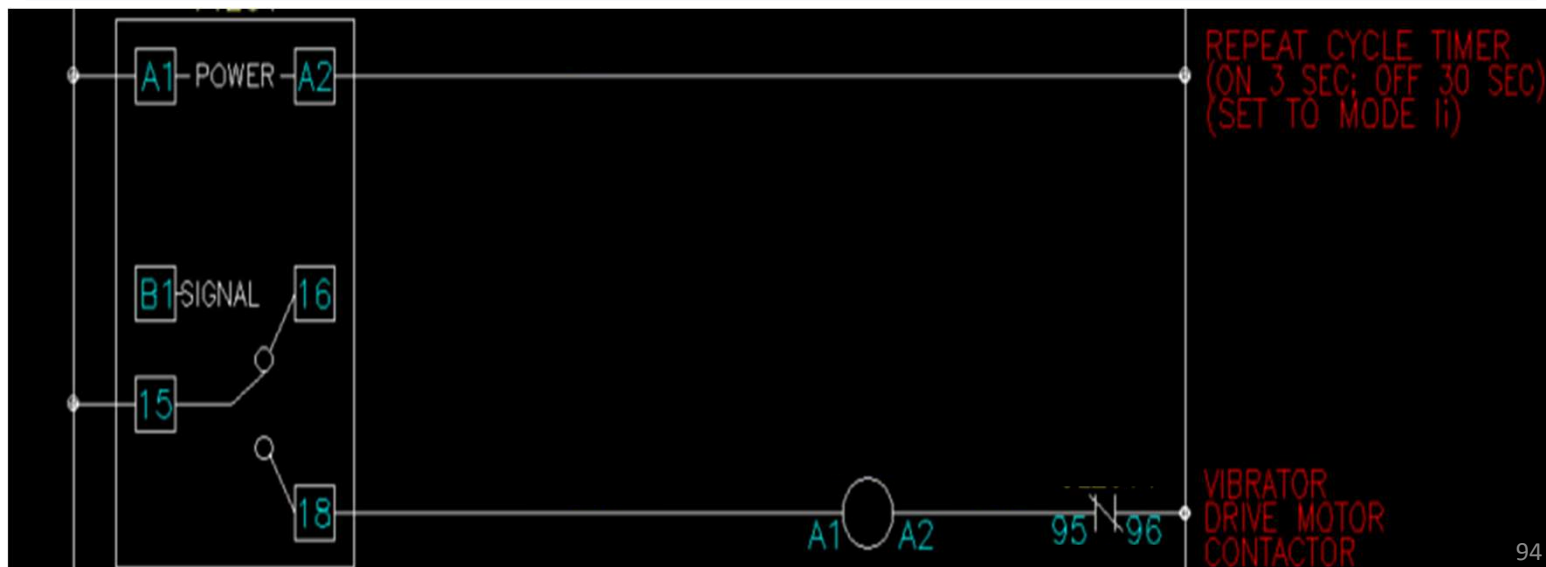
3.

Then the #15 contact on the relay timer closes and sends power through the #18 contact out to the vibrator to turn it on. After the “ON” side of the timer is completed (*factory set to 3 seconds*) the contact will open and the vibrator will turn off. At this point, the “OFF” side of the timer will start timing (*factory set to 30 seconds*). At the completion of the 30 seconds the timer will turn the vibrator on again for 3 seconds.

4.

While the Helix™ is running the vibrator will repeat this cycle until the conveyor is stopped or the material in the Helix™ hopper goes below the low-level sensor.

Details about the ON/OFF Vibrator Cycle Timer Relay can be found on the next page.

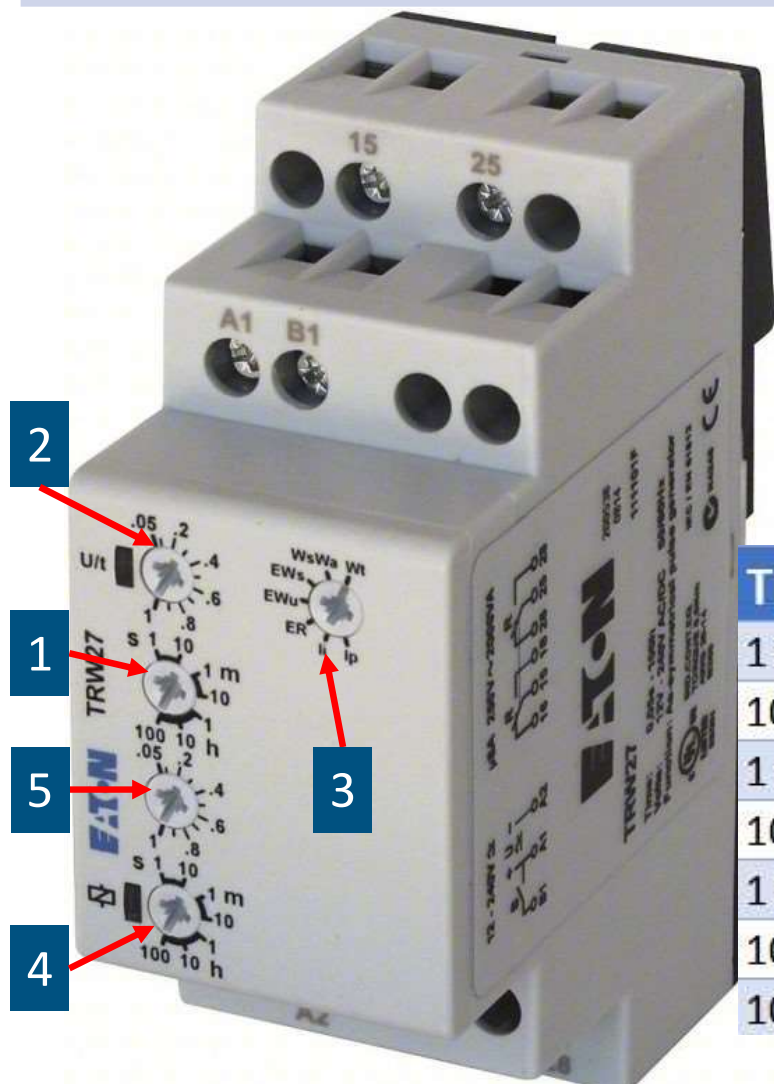


Appendix D: Helix Hopper Vibrator

ON/OFF Vibrator Relay Timer

1. **ON Time Range:** is the base value that determines the amount of adjustment range available.
2. **ON Adjustment Range:** Used to set how long the vibrator will be ON.
3. **Mode Selection:** Dial should be pointing towards “li” for proper operation.
4. **OFF Time Range:** is the base value that determines the amount of adjustment range available.
5. **OFF Adjustment Range:** Used to set how long the vibrator will be OFF

Instructions continued on the next page



If more frequent vibration is needed, start by lowering the off time by small increments.

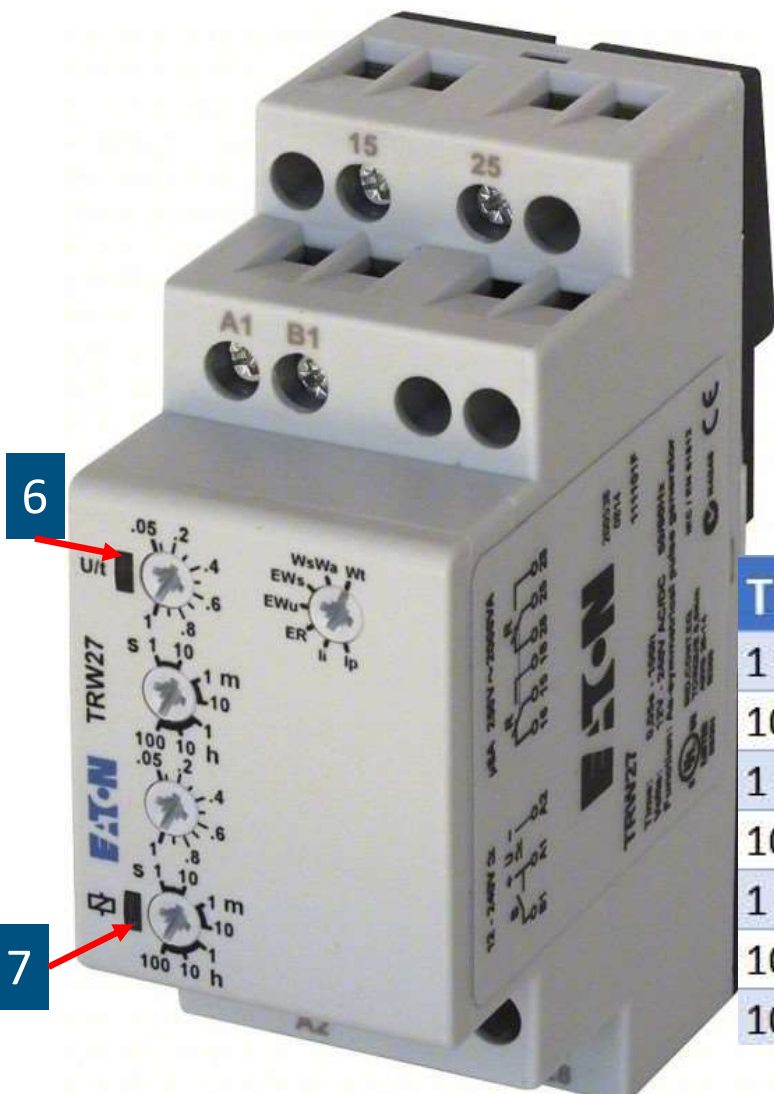
Time Range	Adjustment Range
1 second	50ms to 1 second
10 seconds	500ms to 10 seconds
1 minute	3 seconds to 1 minute
10 minutes	30 seconds to 10 minutes
1 hour	3 minutes to 1 hour
10 hours	30 minutes to 10 hours
100 hours	5 hours to 100 hours

Appendix D: Helix Hopper Vibrator

ON/OFF Vibrator Relay Timer

6. **U/t Green LED:** Once a signal is received then the green light will start flashing. A slow flashing green light indicates it is timing the “ON” time. A fast-flashing green light indicates it is timing the “OFF” time.
7. **Output Indicator Yellow LED:** When the yellow LED comes on it indicates the “ON” signal is being sent to start the vibrator. This light should be illuminated when the vibrator is running.

NOTICE Pneumatic vibrators are controlled the same way. The only difference being that the timer turns on a solenoid to the vibrator instead of an electric motor.



Time Range	Adjustment Range
1 second	50ms to 1 second
10 seconds	500ms to 10 seconds
1 minute	3 seconds to 1 minute
10 minutes	30 seconds to 10 minutes
1 hour	3 minutes to 1 hour
10 hours	30 minutes to 10 hours
100 hours	5 hours to 100 hours

Appendix E: High-Level Discharge Sensor

Hapman Helix™ conveyors come factory equipped with the capability to be interlocked to the equipment we are filling. This feature allows the equipment we are filling to tell the Helix when to start and stop. This appendix details both ways this feature can be utilized.



Prevent serious injury or death.

Electrical installation/modifications must be performed by a qualified electrician and conform to all electrical codes.



Severing and entanglement hazard.

Conveyor can restart automatically with this option

Keep clear of conveyor and all moving parts.

Disconnect and lockout power source before adjusting, cleaning, unclogging or servicing.



Wiring numbers used in the diagrams in this section are **for example only.** **Always refer to your job specific electrical drawings.**

Option 1: High Level Discharge Sensor

A **discharge high level sensor** is defined by Hapman as, any normally closed sensor that is monitoring the high limit fill level of material being discharged from a Hapman Helix™ conveyor. In this instance, it is directly connected to the Hapman supplied control panel.

When utilizing this option, the equipment will operate as such:

When the normally closed discharge high level sensor is installed, and the start button is pressed, the Helix will start right away. The Helix™ will run until the material reaches the discharge high level sensor. The sensor circuit will then open and stop sending a signal to refill. Then the Helix™ will stop. As the downstream process takes material away, the discharge high level sensor will return to a normally closed state. At that time, the conveyor On Delay Restart Timer will start timing and once it elapses, the Helix™ will automatically restart. The conveyor will continuously repeat this cycle until the stop button is pushed or a fault occurs.



The Helix™ can also be paused due to the material level in the Helix hopper dropping below the low-level sensor. Since this is a pause, the unit will restart once the material level in the hopper has been replenished.

Appendix E: High-Level Discharge Sensor

NOTICE

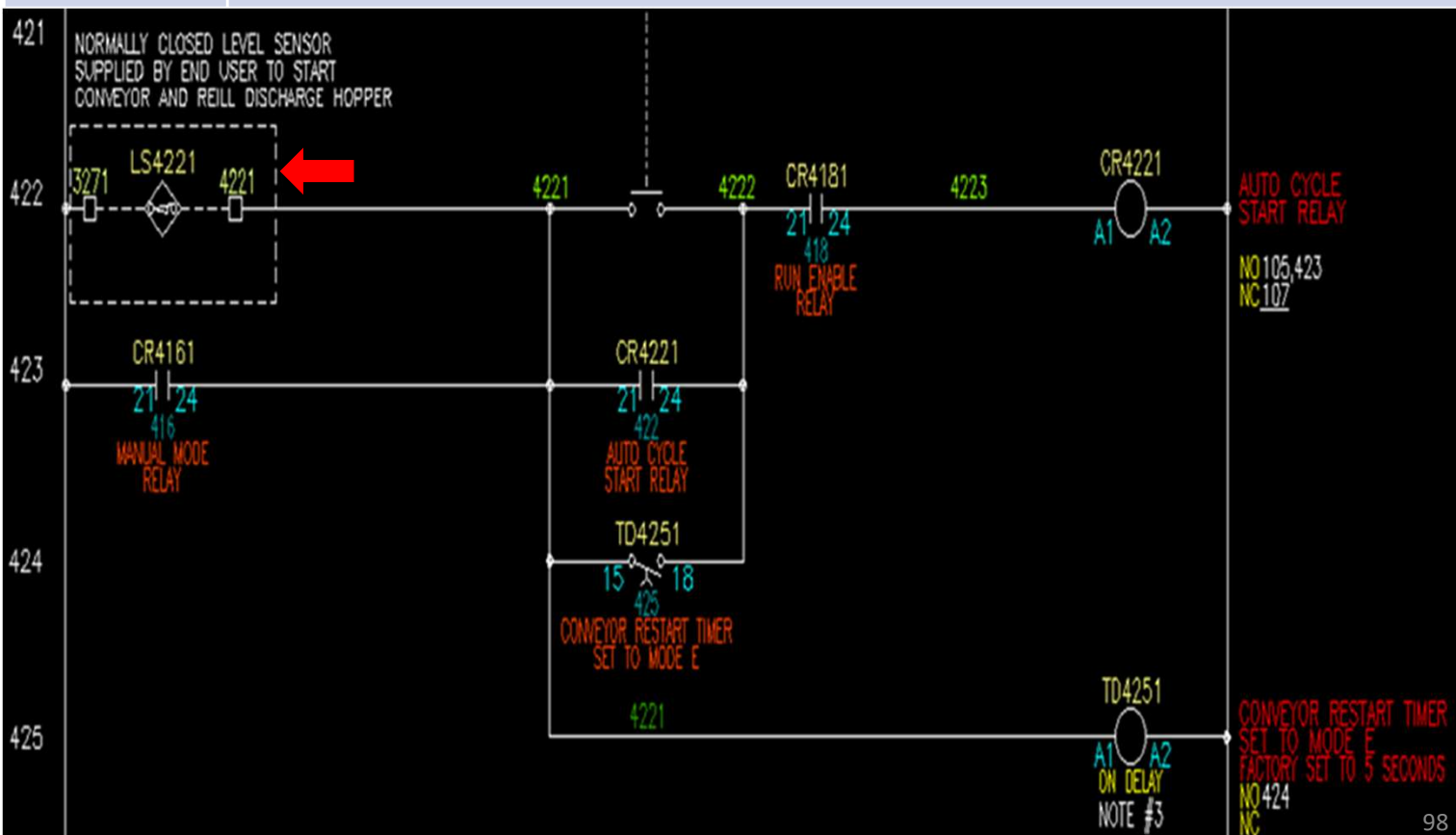
Wiring numbers used in the diagrams below are for example only. **Always** refer to your job specific electrical drawings.



1. Remove the jumper called out between wiring points 3271 and 4221
2. The power from terminal 3271 will power the sensor
3. The signal wiring from the sensor will be inserted into terminal 4221

NOTICE

Below is what the wiring diagram would look like once the jumper has been removed and the sensor wired in.



Appendix E: High-Level Discharge Sensor



Prevent serious injury or death.

Electrical installation/modifications must be performed by a qualified electrician and conform to all electrical codes.

Conveyor Restart On Delay Timer

The Conveyor Restart ON Delay Timer is utilized to prevent the immediate restarting of the Helix™ when the signal from the high-level discharge level sensor or Auto Restart (Call for Material) is received.

This is done for two primary reasons:

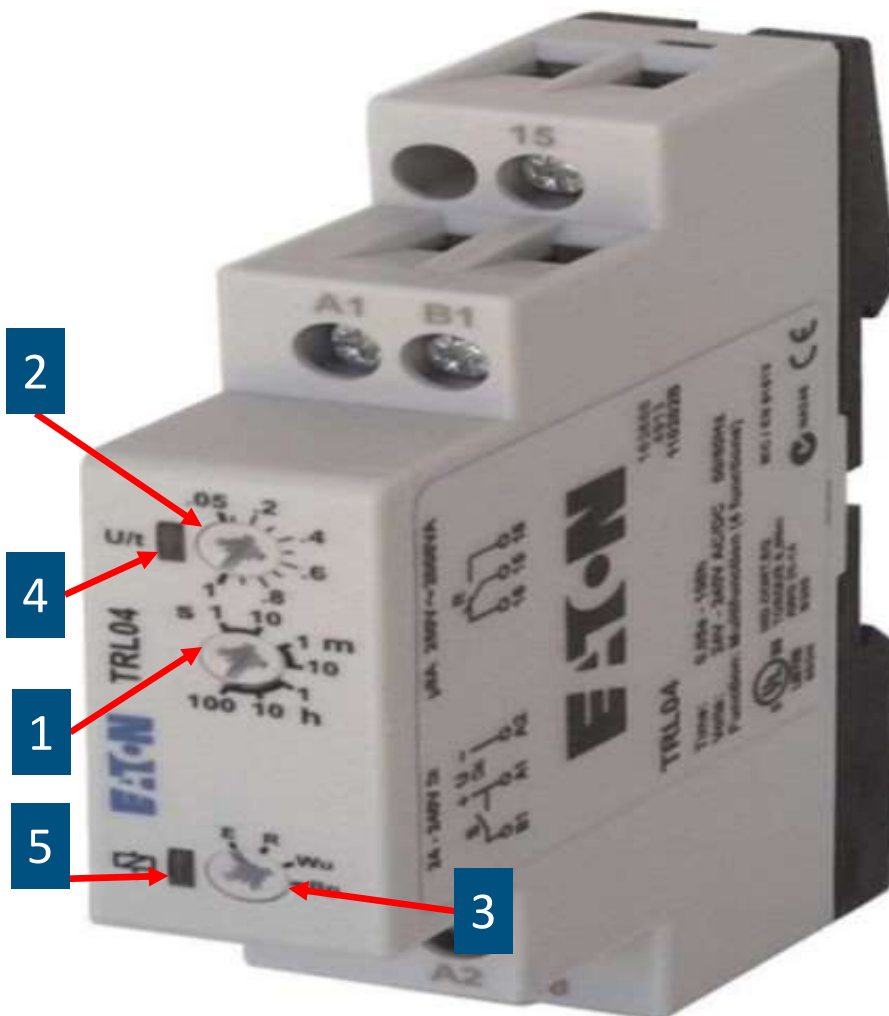
The first is to allow the material to reach the lowest level possible in the vessel being filled without impeding operation. Meaning, that if it takes 15-20 seconds from the point the high-level sensor sends a signal to start the Helix™ and the vessel being filled to go empty, then you can set the refill delay for 10 to 12 seconds to ensure the vessel is always refilled in time.

The second reason for this time is to prevent the Helix™ drive motor from unnecessary short cycling.

The next page describes the mechanical function of the timer and how to adjust time settings.

Appendix E: High-Level Discharge Sensor

1. **Time Range:** is the base value that determines the amount of adjustment range available.
2. **Adjustment Range:** Used to fine tune the exact time frame the conveyor will wait to restart once a signal is received.
3. **Mode Selection:** Dial should be pointing towards “E” for On Delay mode.
4. **U/t Green LED:** This light will be solid green the unit when is idle. Once a signal is received then the green light will start flashing. This indicates the timer is timing.
5. **Output Indicator Yellow LED:** When the yellow LED comes on it indicates the timer is done timing and the Helix should be running.



Time Range	Adjustment Range
1 second	50ms to 1 second
10 seconds	500ms to 10 seconds
1 minute	3 seconds to 1 minute
10 minutes	30 seconds to 10 minutes
1 hour	3 minutes to 1 hour
10 hours	30 minutes to 10 hours
100 hours	5 hours to 100 hours

If an instantaneous start is desired, then you would set the time range to 1 second and the adjustment range to .05

Appendix E-1: Auto Restart/Call for Material

Hapman Helix™ conveyors come factory equipped with the capability to be interlocked to the equipment we are filling. This feature allows the equipment were filling to tell the Helix when to start and stop. This appendix details both ways this feature can be utilized.



Prevent serious injury or death.

Electrical installation/modifications must be performed by a qualified electrician and conform to all electrical codes.



*Wiring numbers used in the diagrams in this section are for example only. **Always refer to your job specific electrical drawings.***

The Auto restart/call for material is defined by Hapman as, isolated normally open contacts in down stream equipment. This relay will utilize Hapman's control power on the dry contacts of the downstream equipment's relay to send a restart signal to the Helix™.

When utilizing this option, the equipment will operate as such:

The operator will hit the start button on the Helix™ conveyor. The Helix™ conveyor will sit idle until it receives a signal/call for material from the downstream equipment it has been interlocked to. Once the signal is received, the Helix™ will start and deliver material until the down stream equipment relay opens and no longer sends a signal. At which point the Helix will stop. This is a continuous process until someone presses the stop button on the control panel, presses the E-stop on the control panel, disengages a safety sensor, or a fault occurs that stops the system. In all instances the start button on the Helix™ control panel would need to be pressed to put the system back into the continuous cycle mode.



The Helix™ can also be paused due to the material level in the Helix hopper dropping below the low-level sensor. Since this is a pause, the unit will restart once the material level in the hopper has been replenished.



This feature is tied to an On Delay Timer that must have the time range set to 1 second and the adjustment range to .05 for proper operation. Reference page 100 for how to set the timer.

Appendix E-1: Auto Restart/Call for Material

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

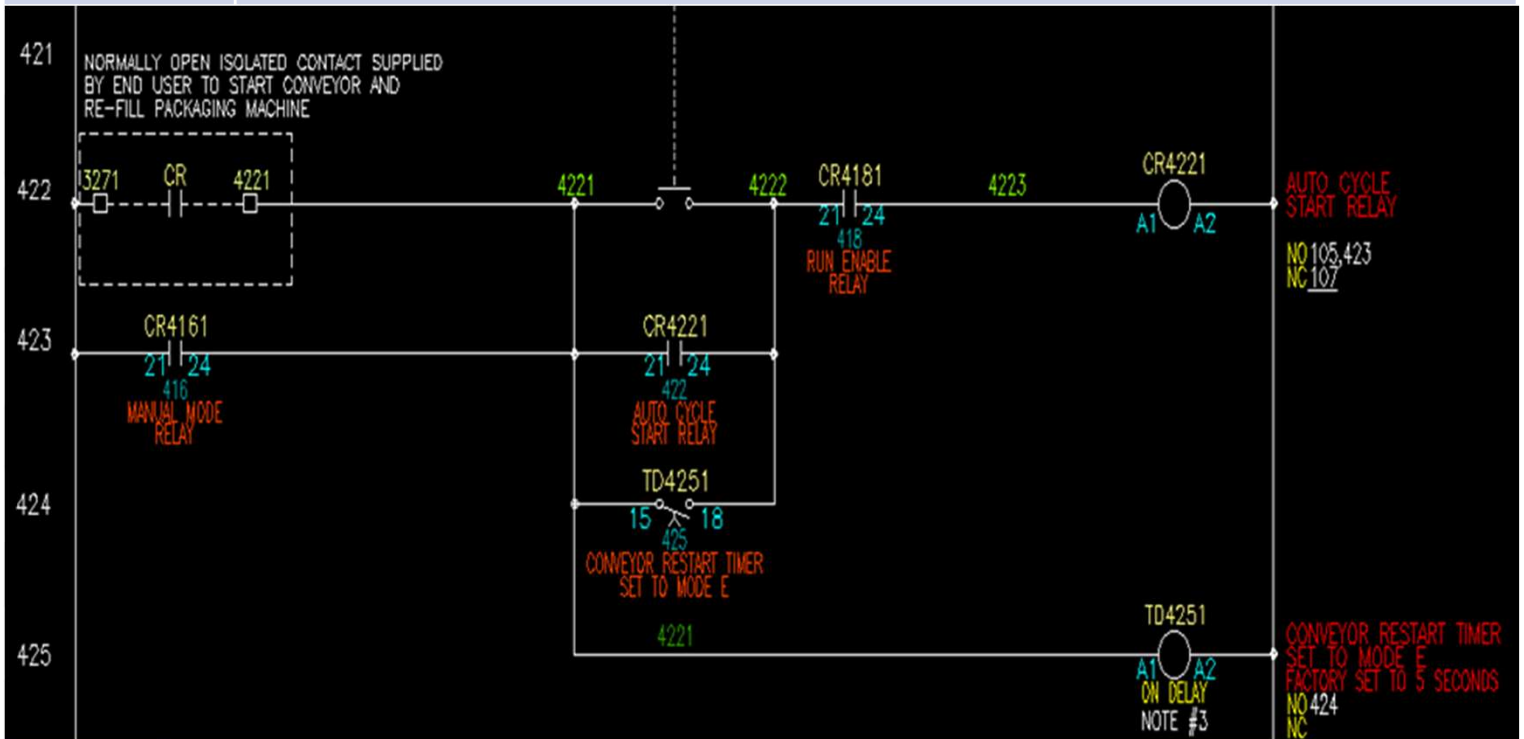


1. Adjust the Conveyor Restart On Delay Timer to 1 second for the time range and .05 seconds for the adjustment range.
Reference page 100 for how to set the timer.
2. Remove the jumper called out between wiring points 3271 and 4221
3. The power from terminal 3271 will be taken to the supply side of a set of dry contacts in the down stream equipment.
4. The signal wire coming back the output side of the dry contacts will land on terminal 4221
5. Once properly wired in, the unit must be put into “Auto Mode” on the selectors switch at the front of the panel.
6. Next, press the start button. The Helix™ will wait for the signal from the down stream equipment. Once the signal is received in the control panel the Helix will turn on.
6. The Helix™ will run until the signal from the down stream equipment is lost. Then the it will stop and wait for the signal again.

Appendix E-1: Auto Restart/Call for Material

NOTICE

*Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.***



Once properly wired the print would look like the image above.

Appendix E-2: Remote Start (Option 1) Operator Required

Hapman Helix™ conveyors come factory equipped with the capability to be started via a dedicated remote start signal. This appendix details the proper controls processes that must be followed to properly and safely implement this feature.



Prevent serious injury or death.
Electrical installation/modifications must be performed by a qualified electrician and conform to all electrical codes.



Severing and entanglement hazard.
Conveyor may start automatically or from a remote location.
Keep clear of conveyor and all moving parts.
Disconnect and lockout power source before adjusting, cleaning, unclogging or servicing.



Wiring numbers used in the diagrams in this section are for example only.
Always refer to your job specific electrical drawings.

The Remote Start (Option 1) is defined by Hapman as a set of isolated normally open contacts on a relay in the customer control panel. This relay will utilize Hapman's control power on the dry contacts of the relay to send a start signal to the Helix™ when the relay is energized. However, operator intervention is still required to initially start the system.

When utilizing this option, the equipment will operate as such:

The operator will press the start button and the unit will sit paused waiting for a remote start signal to be sent to the Helix™ conveyor. The unit will start and stay running until the start signal is removed, the stop button is pressed, an E-stop is pressed, or the unit has a fault. At which time, the Helix™ will stop running. **If the Helix™ is stopped due to an E-stop or fault the remote start signal must be removed!**



Follow the instructions in the rest of this section to make sure this option is implemented per Hapman's best practices.
Failure to do so could result in serious or fatal injury.

Appendix E-2: Remote Start (Option 1)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

When implementing the remote start option, there are Hapman supplied interlocks that must be monitored by the system sending the remote start signal. These are easily identifiable in the panel by the orange wires on the terminal strip. These orange wires are the ones that will be utilized for monitoring the interlocks.

1. **Emergency Stop Push Button:** This set of normally closed dry contacts will be utilized to know when an E-Stop has been pressed on our control panel.
2. **Hopper Low-Level Sensor:** This set of normally open dry contacts is utilized to know when the material level in the hopper has dropped below the sensor and the unit is being paused until the material level has been refilled above the level sensor.
3. **Operating Mode Selector Switch:** This set of normally open dry contacts is utilized to know if the unit is in Manual, Auto, or Off modes.
4. **Helix Conveyor Running Relay:** This set of normally open dry contacts is utilized to know when the Helix™ conveyor drive is running.

WARNING

All the interlocks identified above should be utilized and monitored when implementing the remote start option. Failure to do so could lead to damage to the equipment or injury to operators.



Appendix E-2: Remote Start (Option 1)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

WARNING

Conveyor can start automatically from a remote location, without warning, with this option.
Severing and entanglement hazard.

1. **Emergency Stop Push Button (E-stop):** This set of normally closed dry contacts will be utilized to know when an E-Stop has been pressed on our control panel.

To properly wire this interlock, you would need to bring control power (24vdc, 120vac, etc.) from the remote panel to the Hapman panel and land it on the proper terminal strip location for the Emergency Stop button. For this example, the remote panel control power wire would land on terminal 5011 on the terminal strip.

Then a second wire would be run from terminal 5012 in the Hapman panel back to the remote panel. If the E-Stop push button has not been pressed or the Hapman panel powered down, you will be receiving your control voltage back indicating the E-stop circuit is intact. If the E-stop is pressed you would lose the signal.

Whenever an E-stop is pressed on the Hapman panel the remote run signal being sent must be removed. The remote start signal should not be reinstated until there has been visual confirmation that the issue that led to the E-stop being pressed has been resolved and the safety circuit has been restored.



Appendix E-2: Remote Start (Option 1)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

When implementing the remote start option, there are Hapman supplied interlocks that must be monitored by the system sending the remote start signal. These are easily identifiable in the panel by the orange wires on the terminal strip. These orange wires are the ones that will be utilized for monitoring the interlocks.

2. **Hopper Low-Level Sensor:** This set of normally open dry contacts is utilized to know when the material level in the hopper has dropped below the sensor and the unit is being paused until the material level has been refilled above the level sensor.

To properly wire this interlock, you would need to bring control power (24vdc, 120vac, etc.) from the remote panel to the Hapman panel and land it on the proper terminal strip location for the low-level sensor. For this example, the remote panel control power wire would land on terminal 5013 on the terminal strip.

Then a second wire would be run from terminal 5014 in the Hapman panel back to the remote panel. If the low-level sensor is seeing material, you will be receiving your control voltage back.

Whenever the low-level signal is lost the remote run signal being sent must be removed.
The remote start signal should not be reinstated until the low-level sensor signal has been seen for 5 seconds.



Appendix E-2: Remote Start (Option 1)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

When implementing the remote start option, there are Hapman supplied interlocks that must be monitored by the system sending the remote start signal. These are easily identifiable in the panel by the orange wires on the terminal strip. These orange wires are the ones that will be utilized for monitoring the interlocks.

3. **Operating Mode Selector Switch:** This set of normally open dry contacts is utilized to know if the unit is in Manual, Auto, or Off modes.

To properly wire this interlock, you would need to bring control power (24vdc, 120vac, etc.) from the remote panel to the Hapman panel and land it on the proper terminal strip location for the low-level sensor. For this example, the remote panel control power wire would land on terminal 5015 on the terminal strip.

Then a second wire would be run from terminal 5016 in the Hapman panel back to the remote panel. If the low-level sensor is seeing material, you will be receiving your control voltage back.

Whenever the unit is not in “Auto” the remote run signal being sent must be removed.

The remote start signal should not be reinstated until there has been confirmation that the reason that led to the unit being taken out of “Auto” has been resolved and the selector switch has been put back into the “Auto” position.



Appendix E-2: Remote Start (Option 1)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

When implementing the remote start option, there are Hapman supplied interlocks that must be monitored by the system sending the remote start signal. These are easily identifiable in the panel by the orange wires on the terminal strip. These orange wires are the ones that will be utilized for monitoring the interlocks.

4. **Helix™ Conveyor Running Relay:** This set of normally open dry contacts is utilized to know when the Helix™ conveyor drive is running.

To properly wire this interlock, you would need to bring control power (24vdc, 120vac, etc.) from the remote panel to the Hapman panel and land it on the proper terminal strip location for the low-level sensor. For this example, the remote panel control power wire would land on terminal 5031 on the terminal strip.

Then a second wire would be run from terminal 5032 in the Hapman panel back to the remote panel. When the Helix™ drive motor starts you will receive your control voltage back. Indicating the Helix™ is now running.

Whenever the Helix™ running relay is not seen within 5 seconds of being told to run the remote run signal must be removed. The remote start signal should not be reinstated until there has been an investigation as to why the unit is not starting when told. Once resolved and the all clear is given then the remote run signal can be reinstated.



Appendix E-2: Remote Start (Option 1)

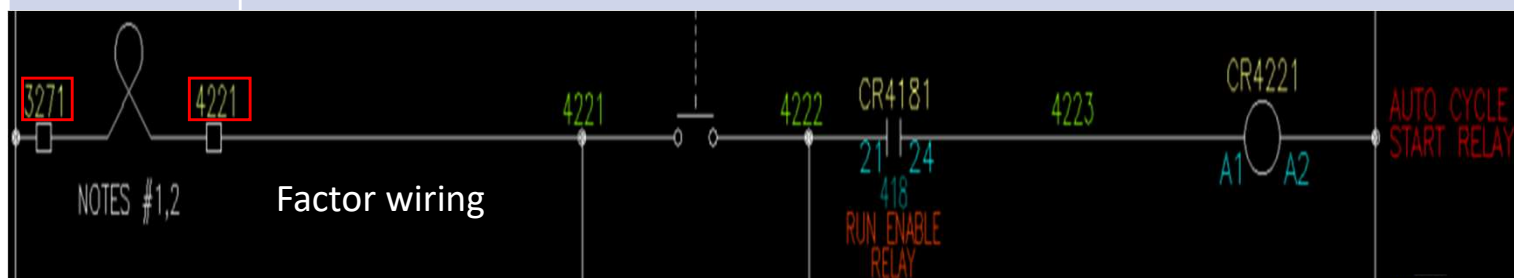
NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

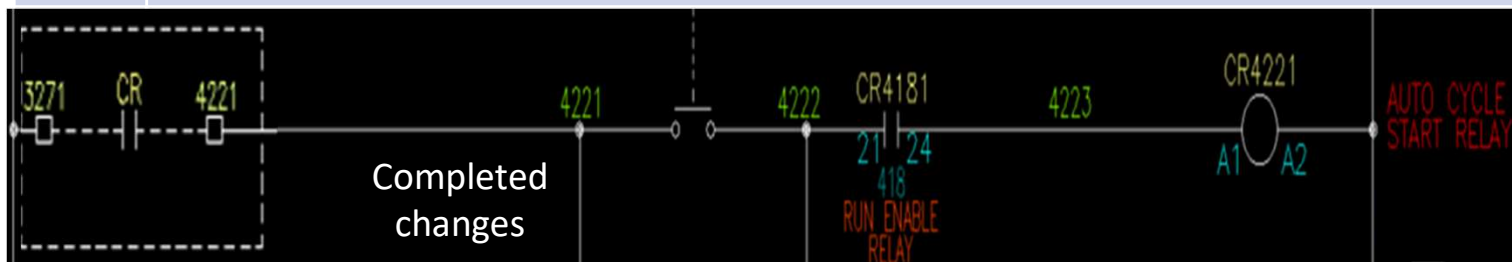
When implementing the remote start option, there are two Hapman supplied wire jumpers on the terminal strip in the panel. One jumper must be removed and the other one relocated before the option can be utilized.

WARNING

All the interlocks identified on the previous pages should be wired and ready to test before testing the remote start option. Failure to do so could lead to damage to the equipment or injury to operators.



1. Remove the jumper called out between wiring points 3271 and 4221
2. The power from terminal 3271 will be taken back to an isolated set of normally open dry contacts in the remote panel being utilized for the remote start signal.
3. The power will be sent back to the Hapman via wire 4221 when the normally open dry contacts close in the remote panel.



NOTICE

This feature is tied to an On Delay Timer that must have the time range set to 1 second and the adjustment range to .05 for proper operation. Reference page 100 for instructions.

Instructions for the relocating the next jumper required for the remote start can be found on the next page.

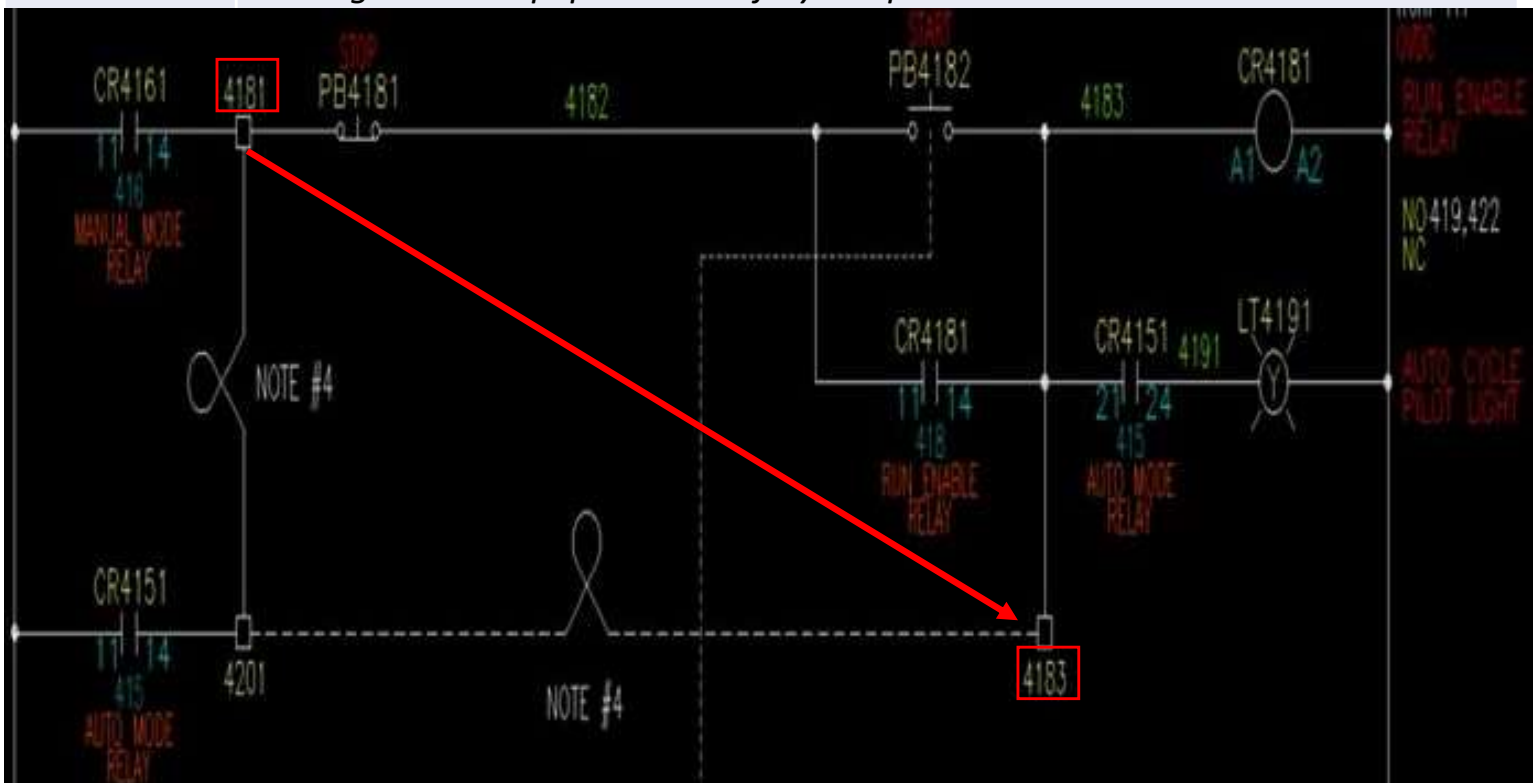
Appendix E-2: Remote Start (Option 1)

NOTICE

*Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.***

⚠ WARNING

All the interlocks identified on the previous pages should be wired and ready to test before testing the remote start option. Failure to do so could lead to damage to the equipment or injury to operators.



1. Remove one side of the jumper from terminal 4181
2. Place the wire in terminal 4183

NOTICE

This feature is tied to an On Delay Timer that must have the time range set to 1 second and the adjustment range to .05 for proper operation. Reference page 100 for instructions.

Appendix E-3: Remote Start (Option 2) No Operator Required

Hapman Helix™ conveyors come factory equipped with the capability to be started via a dedicated remote start signal. This appendix details the proper controls processes that must be followed to properly and safely implement this feature.



Prevent serious injury or death.
Electrical installation/modifications must be performed by a qualified electrician and conform to all electrical codes.



Severing and entanglement hazard.
Conveyor may start automatically or from a remote location.
Keep clear of conveyor and all moving parts.
Disconnect and lockout power source before adjusting, cleaning, unclogging or servicing.



Conveyor will start automatically from a remote location with this option.
Severing and entanglement hazard.



Wiring numbers used in the diagrams in this section are for example only.
Always refer to your job specific electrical drawings.

The Remote Start (Option 2) is defined by Hapman as a set of isolated normally open contacts on a relay in the customer control panel. This relay will utilize Hapman's control power on the dry contacts of the relay to send a start signal to the Helix™ when the relay is energized. **This option eliminates the operator's ability to start or stop the equipment in "Auto" mode locally.**

When utilizing this option, the equipment will operate as such:

A remote start signal will be sent to the Helix™ conveyor. The unit will start and stay running until the start signal is removed, an E-stop is pressed, or the unit has a fault. At which time, the Helix™ will stop running. **If the Helix™ is stopped due to an E-stop or fault the remote start signal must be removed!**



Follow the instructions in the rest of this section to make sure this option is implemented per Hapman's best practices. Failure to do so could result in serious or fatal injury.

Appendix E-3: Remote Start (Option 2)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

When implementing the remote start option, there are Hapman supplied interlocks that must be monitored by the system sending the remote start signal. These are easily identifiable in the panel by the orange wires on the terminal strip. These orange wires are the ones that will be utilized for monitoring the interlocks.

1. **Emergency Stop Push Button:** This set of normally closed dry contacts will be utilized to know when an E-Stop has been pressed on our control panel.
2. **Hopper Low-Level Sensor:** This set of normally open dry contacts is utilized to know when the material level in the hopper has dropped below the sensor and the unit is being paused until the material level has been refilled above the level sensor.
3. **Operating Mode Selector Switch:** This set of normally open dry contacts is utilized to know if the unit is in Manual, Auto, or Off modes.
4. **Helix Conveyor Running Relay:** This set of normally open dry contacts is utilized to know when the Helix™ conveyor drive is running.

WARNING

All the interlocks identified above should be utilized and monitored when implementing the remote start option. Failure to do so could lead to damage to the equipment or injury to operators.



Appendix E-3: Remote Start (Option 2)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

⚠ DANGER

Conveyor **will start** automatically from a remote location with this option. Severing and entanglement hazard.

1. **Emergency Stop Push Button (E-stop):** This set of normally closed dry contacts will be utilized to know when an E-Stop has been pressed on our control panel.

To properly wire this interlock, you would need to bring control power (24vdc, 120vac, etc.) from the remote panel to the Hapman panel and land it on the proper terminal strip location for the Emergency Stop button. For this example, the remote panel control power wire would land on terminal 5011 on the terminal strip.

Then a second wire would be run from terminal 5012 in the Hapman panel back to the remote panel. If the E-Stop push button has not been pressed or the Hapman panel powered down, you will be receiving your control voltage back indicating the E-stop circuit is intact.

Whenever an E-stop is pressed on the Hapman panel the remote run signal being sent must be removed! The remote start signal should not be reinstated until there has been visual confirmation that the issue that led to the E-stop being pressed has been resolved and the safety circuit has been restored.



Appendix E-3: Remote Start (Option 2)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

When implementing the remote start option, there are Hapman supplied interlocks that must be monitored by the system sending the remote start signal. These are easily identifiable in the panel by the orange wires on the terminal strip. These orange wires are the ones that will be utilized for monitoring the interlocks.

2. **Hopper Low-Level Sensor:** This set of normally open dry contacts is utilized to know when the material level in the hopper has dropped below the sensor and the unit is being paused until the material level has been refilled above the level sensor.

To properly wire this interlock, you would need to bring control power (24vdc, 120vac, etc.) from the remote panel to the Hapman panel and land it on the proper terminal strip location for the low-level sensor. For this example, the remote panel control power wire would land on terminal 5013 on the terminal strip.

Then a second wire would be run from terminal 5014 in the Hapman panel back to the remote panel. If the low-level sensor is seeing material, you will be receiving your control voltage back.

Whenever the low-level signal is lost the remote run signal being sent must be removed. The remote start signal should not be reinstated until the low-level sensor signal has been seen for 5 seconds.



Appendix E-3: Remote Start (Option 2)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

When implementing the remote start option, there are Hapman supplied interlocks that must be monitored by the system sending the remote start signal. These are easily identifiable in the panel by the orange wires on the terminal strip. These orange wires are the ones that will be utilized for monitoring the interlocks.

3. **Operating Mode Selector Switch:** This set of normally open dry contacts is utilized to know if the unit is in Manual, Auto, or Off modes.

To properly wire this interlock, you would need to bring control power (24vdc, 120vac, etc.) from the remote panel to the Hapman panel and land it on the proper terminal strip location for the low-level sensor. For this example, the remote panel control power wire would land on terminal 5015 on the terminal strip.

Then a second wire would be run from terminal 5016 in the Hapman panel back to the remote panel. If the low-level sensor is seeing material, you will be receiving your control voltage back.

Whenever the unit is not in “Auto” the remote run signal being sent must be removed.

The remote start signal should not be reinstated until there has been confirmation that the reason that led to the unit being taken out of “Auto” has been resolved and the selector switch has been put back into the “Auto” position.



Appendix E-3: Remote Start (Option 2)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

When implementing the remote start option, there are Hapman supplied interlocks that must be monitored by the system sending the remote start signal. These are easily identifiable in the panel by the orange wires on the terminal strip. These orange wires are the ones that will be utilized for monitoring the interlocks.

4. **Helix™ Conveyor Running Relay:** This set of normally open dry contacts is utilized to know when the Helix™ conveyor drive is running.

To properly wire this interlock, you would need to bring control power (24vdc, 120vac, etc.) from the remote panel to the Hapman panel and land it on the proper terminal strip location for the low-level sensor. For this example, the remote panel control power wire would land on terminal 5031 on the terminal strip.

Then a second wire would be run from terminal 5032 in the Hapman panel back to the remote panel. When the Helix™ drive motor starts you will receive your control voltage back. Indicating the Helix is now running.

Whenever the Helix™ running relay is not seen within 5 seconds of being told to run the remote run signal being sent must be removed. The remote start signal should not be reinstated until there has been an investigation as to why the unit is not starting when told. Once resolved and the all clear is given then the remote run signal can be reinstated.



Appendix E-3: Remote Start (Option 2)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always** refer to your job specific electrical drawings.



1. Verify the jumper between 3271 and 4221 is in place on the terminal strip.

NOTICE

This jumper must remain in place or Remote Start (Option 2) will not function as intended.

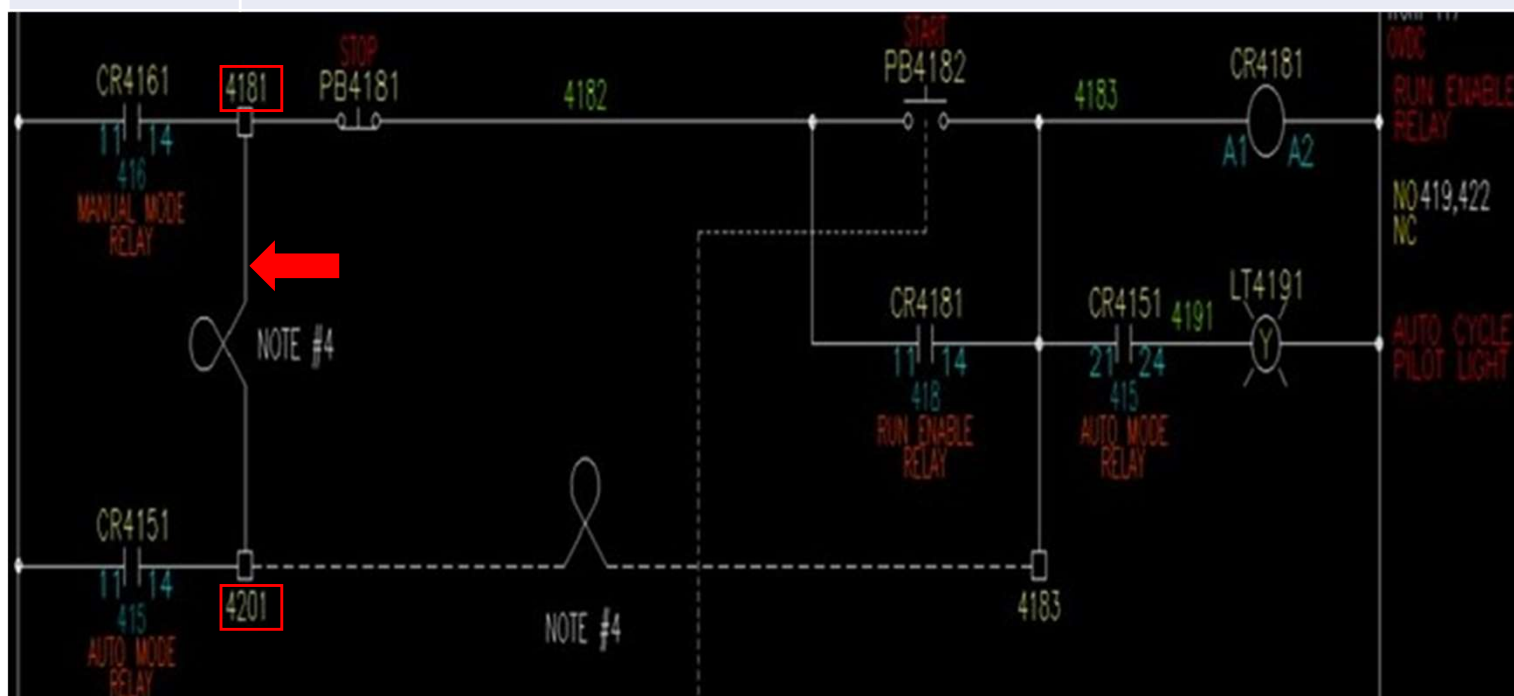
Appendix E-3: Remote Start (Option 2)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always refer to your job specific electrical drawings.**

NOTICE

All the interlocks identified in this section must be wired and ready to test before testing the remote start option. Failure to do so could lead to damage of the equipment or injury to operators.



2. Remove the wire jumper between terminals 4181 and 4201.
3. For this option, the jumper will be removed completely and not relocated as indicated by the dotted lines.

NOTICE

This feature is tied to an On Delay Timer that must have the time range set to 1 second and the adjustment range to .05 for proper operation. Reference page 100 for instructions.

WARNING

This option will eliminate the operator's ability to start and stop the machine locally when the unit is in "Auto" mode.

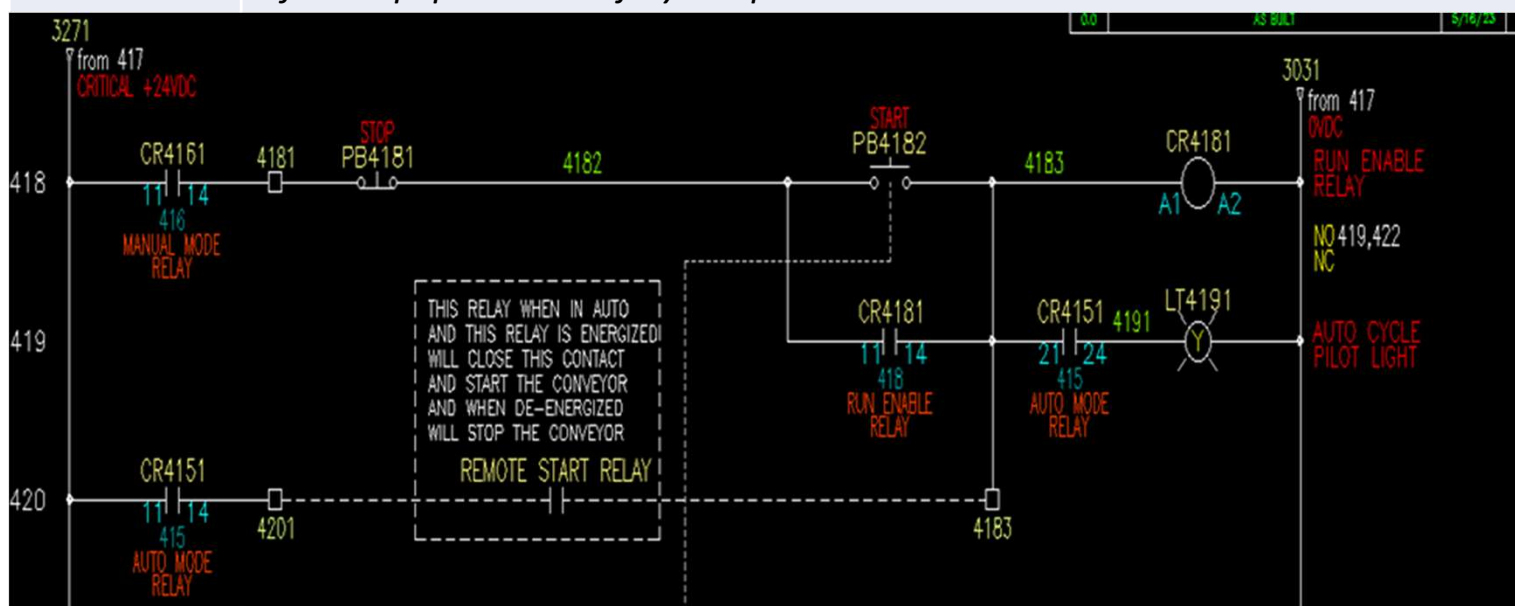
Appendix E-3: Remote Start (Option 2)

NOTICE

Wiring numbers used in the diagrams below are for example only. **Always** refer to your job specific electrical drawings.

NOTICE

All the interlocks identified in this section must be wired and ready to test before testing the remote start option. Failure to do so could lead to damage of the equipment or injury to operators.



-
2. Run a wire from 4201 on the terminal strip to the supply side of the dry contact on the remote start relay in the customer panel.
3. Bring the wire from the outgoing side of the dry contact to the Hapman panel and land on terminal 4183.

NOTICE

This feature is tied to an On Delay Timer that must have the time range set to 1 second and the adjustment range to .05 for proper operation. Reference page 100 for instructions.

⚠ WARNING

This option will eliminate the operator's ability to start and stop the machine locally when the unit is in "Auto" mode.

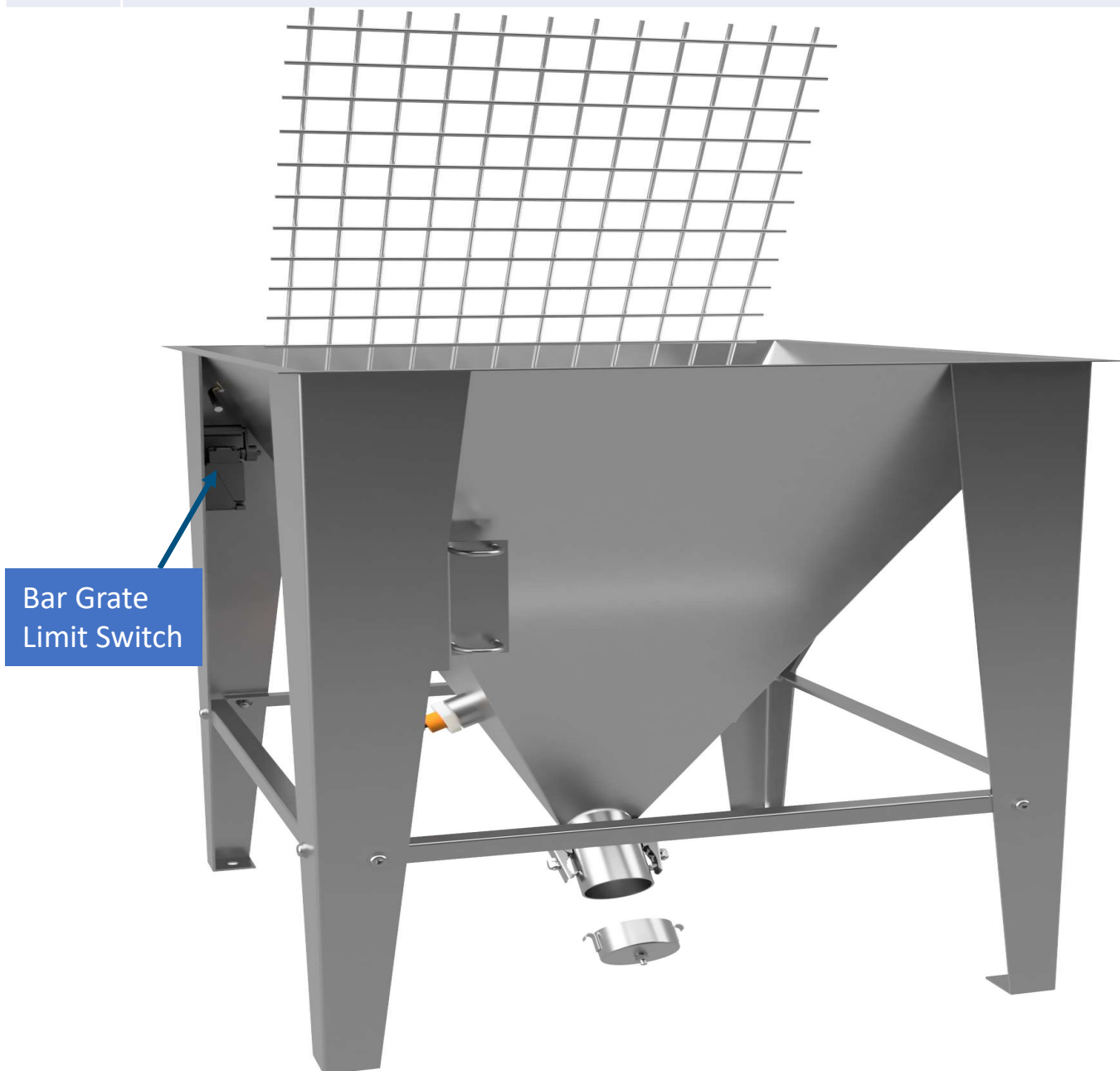
! DANGER

**Conveyor will start automatically from a remote location with this option.
Severing and entanglement hazard.**

Appendix F: Mechanical Interlock on Hinged Gate Helix Hopper

Hapman's standard hinged bar gate hoppers incorporate a mechanical interlock limit switch. It acts as an interlock to prevent the Helix™ conveyor from running when the limit switch is not engaged.

1. The mechanical interlock/bar gate limit switch is normally found inside either the left or right front leg on the hopper



Appendix F: Mechanical Interlock on Hinged Gate Helix Hopper



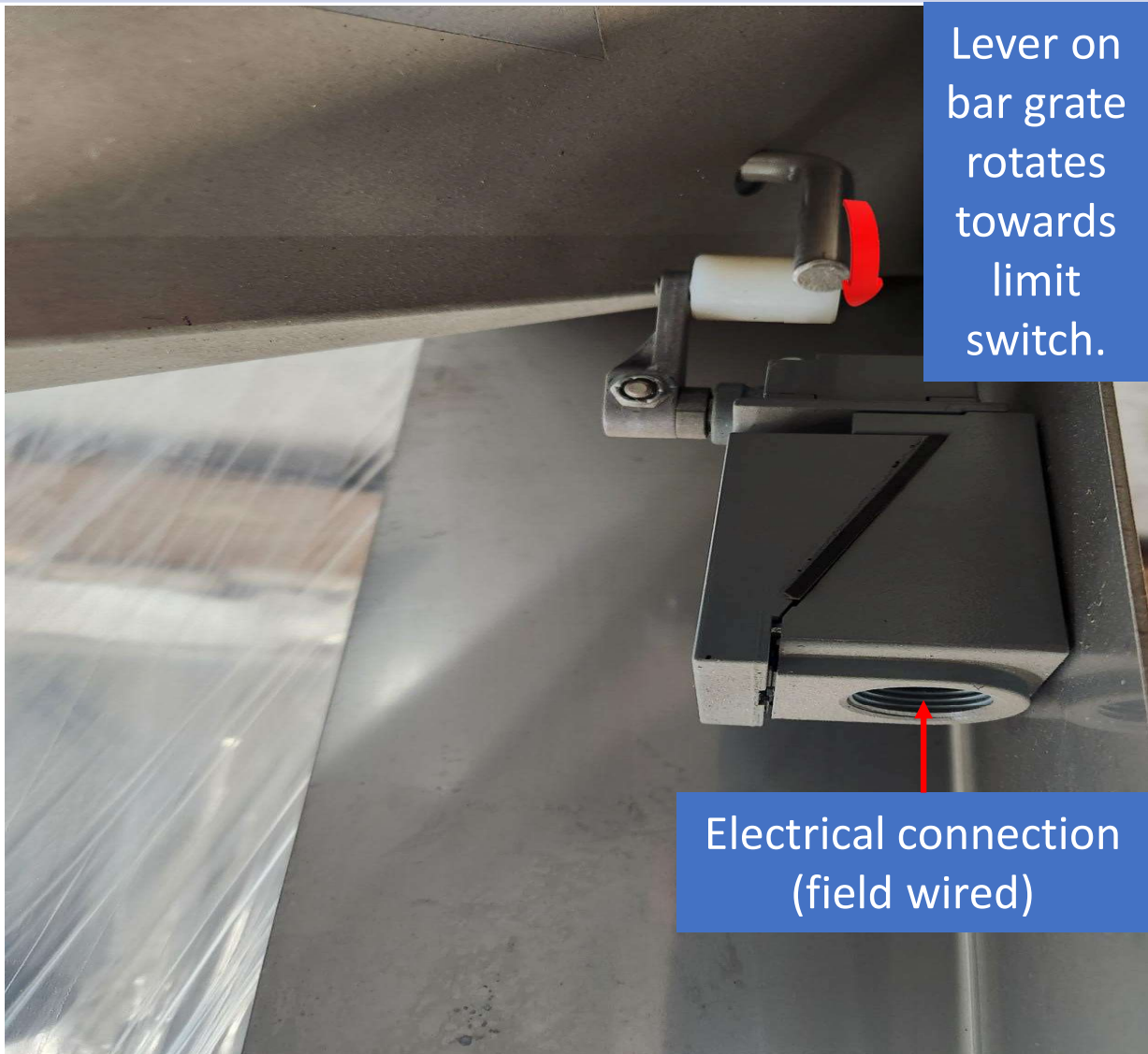
Prevent serious injury or death.

Electrical service installation must be performed by a qualified electrician and conform to all electrical codes.

Hapman's hinged bar grate hoppers incorporate a mechanical interlock limit switch. It acts as an interlock to prevent the Helix™ conveyor from running when the limit switch is not engaged.

2.

The lever on the bar grate extends out to engage the limit switch when the bar grate is in the closed position. This completes a circuit that allows control power to be supplied to the rest of the Helix™ control panel.



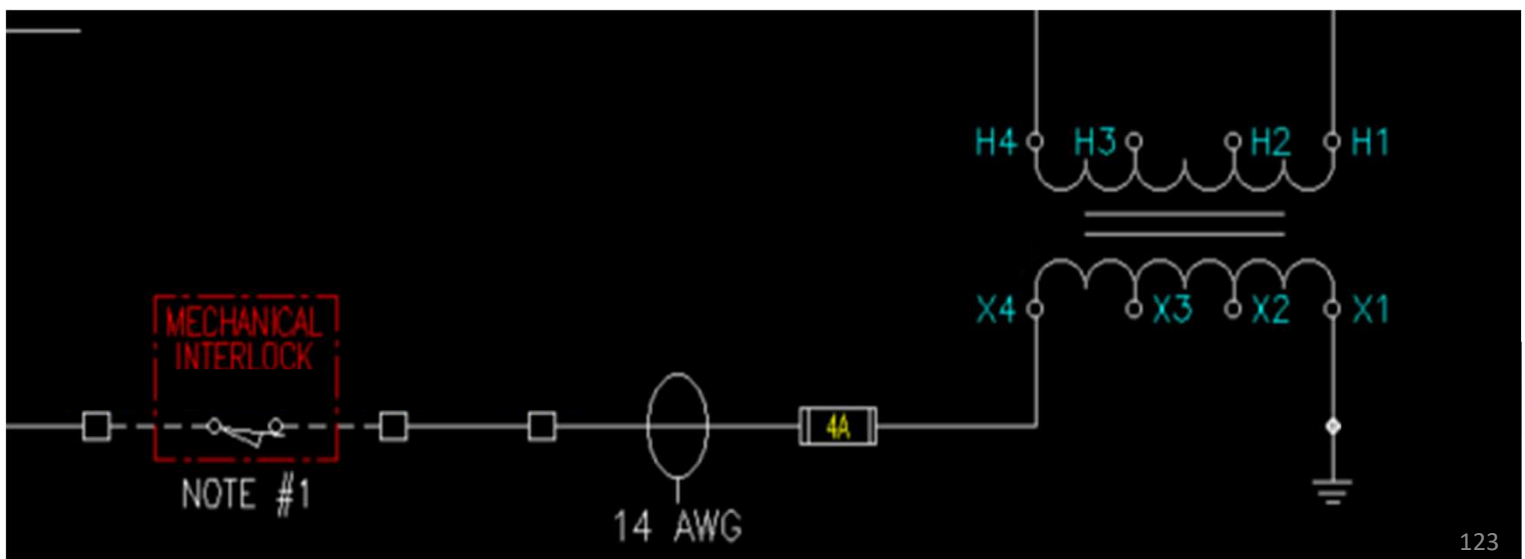
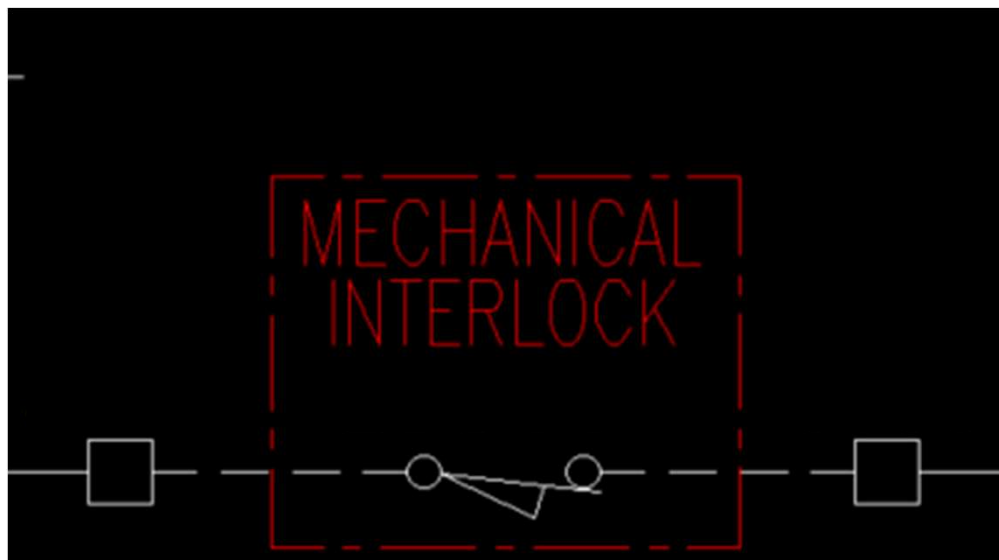
Lever on
bar grate
rotates
towards
limit
switch.

Electrical connection
(field wired)

Appendix F: Mechanical Interlock on Hinged Gate Helix Hopper

Hapman's hinged bar grate hoppers incorporate a mechanical interlock limit switch that prevents the Helix™ conveyor from running when the limit switch is not engaged.

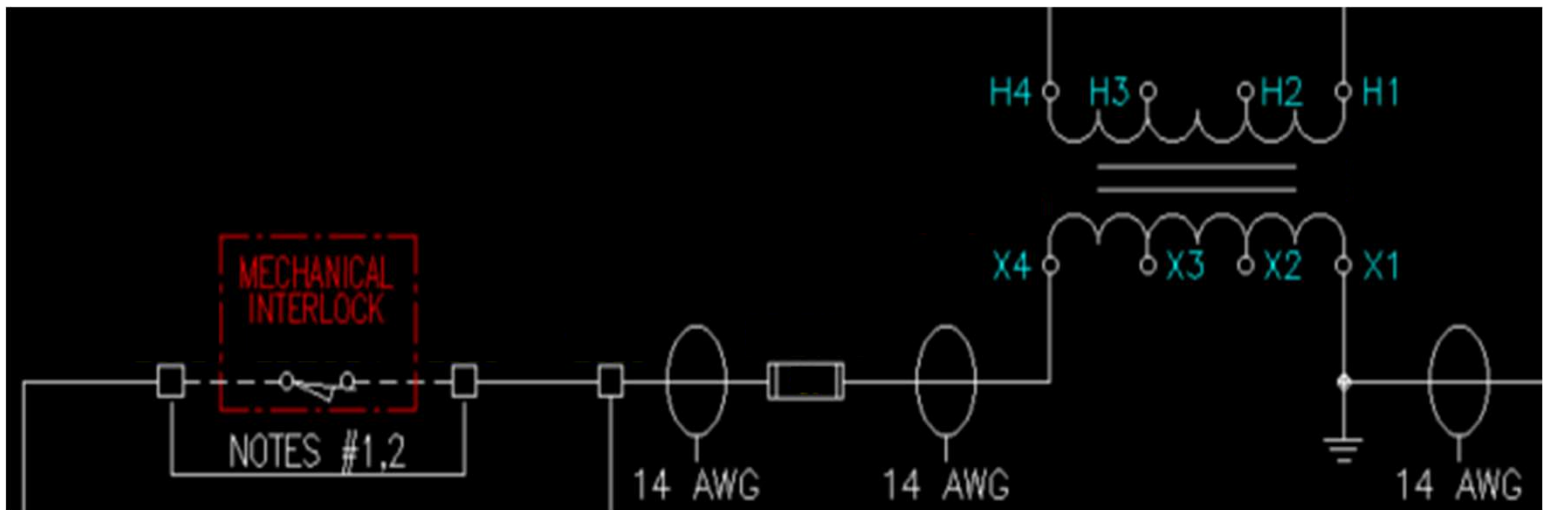
3. If Hapman provided controls for the unit then you can reference our electrical prints for where to wire the mechanical interlock/limit switch into the control panel. Below are reference images for how we call out this wiring point.
4. The sensor is wired as "Normally Open" and held closed when the bar grate is closed.



Appendix F: Mechanical Interlock on Hinged Gate Helix Hopper

Not all Helix™ conveyors have mechanical interlocks. While the terminals are present in all Helix panels, we place a jumper around the interlock from the factory if a safety interlock device is being utilized instead. Reference your electrical drawings or contact Hapman Customer Service with any questions.

Below is an example of what this interlock would look like on Hapman supplied electrical prints if it is not be utilized.



Appendix G: Steel Casing Installation

Some Helix™ conveyors are equipped with steel casings instead of the standard poly casing. The installation procedure is slightly different. The largest differences being that the steel casing is not flexible so it must be installed accurately, and the compression couplings do not have a stepped gasket.

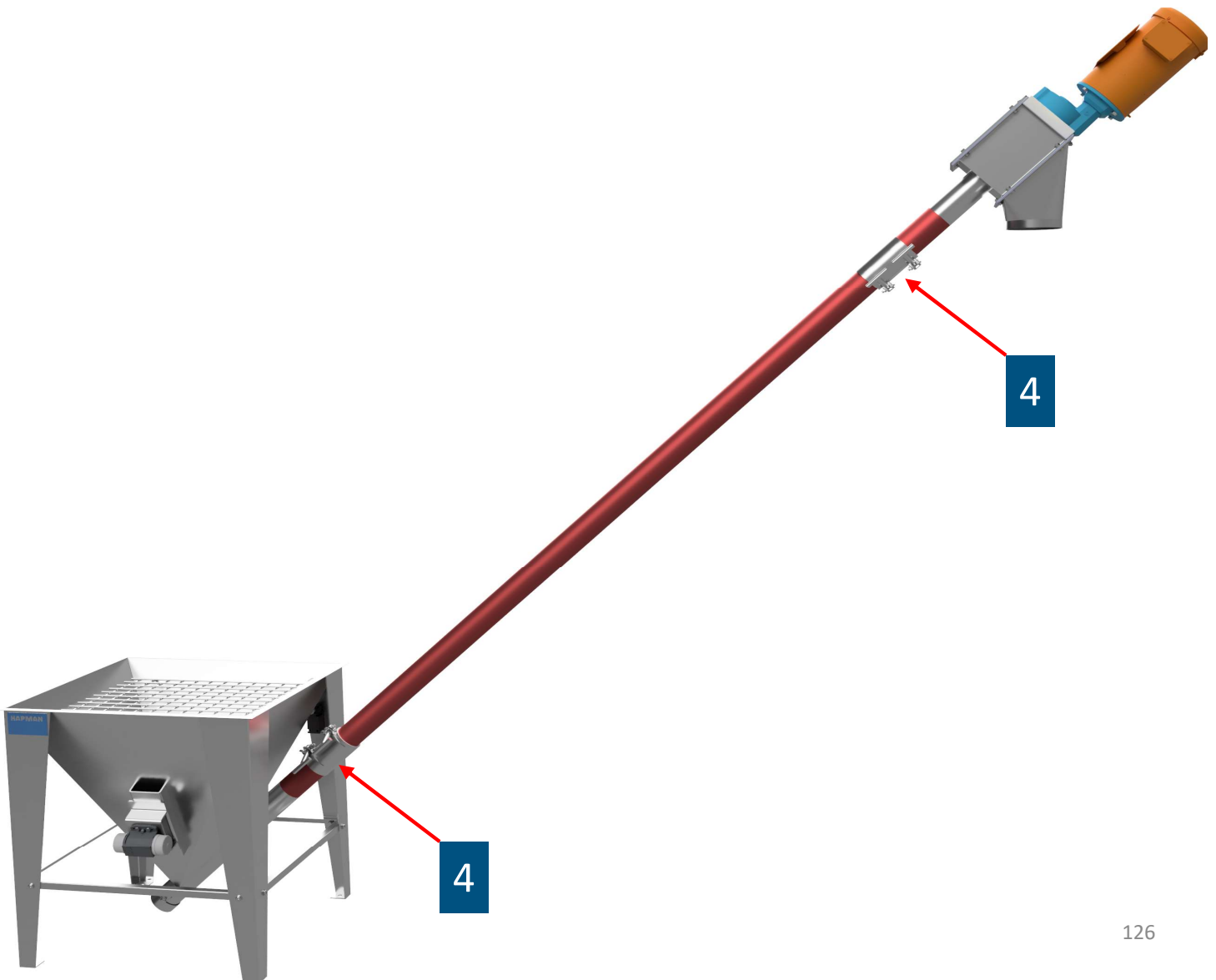
Straight Gasket Compression Coupling

The straight gasket design on the compression couplings are utilized when the standard poly casing has been replaced with a steel casing. This design allows you to secure the Hapman supplied steel casing to the inlet on the Helix™ drive box and hopper boot of the same diameter.



Appendix G: Steel Casing Installation

1. Perform the installation pre-checks as called out on page 21 of this manual.
2. Install the drive section of the Helix™ as called out on pages 22-24 of this manual.
3. Inspect casing to make sure ends are cut square and straight.
4. Slide the straight compression couplings onto the casing and slide down a few inches from both ends of the steel casing.



Appendix G: Steel Casing Installation

5. Verify that the casing butts up square and flush to both the drive and hopper inlets
6. Slide the straight compression couplings into place. Half the coupling on the casing the other on the inlets.
7. Tighten the compression couplings down and verify the casing is aligned straight.

NOTICE

Steel casings can require additional support in the middle due to the added weight of the casing. Reference page 25 for examples of casing supports.

NOTICE

For series 250 Helix™ units, the casing will slide into the hopper and drive inlets just like a standard poly casing. However, the difference between the OD of the casing and the ID of the inlets is so nominal that a straight coupler is utilized instead of a stepped coupler.

